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INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**
Course Code: **MLA0204**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Max Marks: **50 Marks**
Date : 31/10/26

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.30	0.10

Analyse the distribution by (a) finding the marginals of X and Y, (b) computing E[X], Var(X) and E[Y], (c) computing P(X = 2 | Y = 1), (d) computing Cov(X, Y) and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification.

(6) (CO1) (BL4)

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.)

(4) (CO1) (BL4)

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality.

(3) (CO1) (BL4)

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute KL(P||Q) and KL(Q||P) in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss.

(3) (CO1) (BL4)

Task 7. Two binary variables A and B have the joint distribution P(0,0) = 0.4, P(0,1) = 0.1, P(1,0) = 0.1, P(1,1) = 0.4. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A.

(4) (CO1) (BL4)

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x*, classify x = 4.5, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases.

(4) (CO1) (BL4)

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal.

(3) (CO1) (BL5)

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.5	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of θ. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a false alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

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ML2021L - FOML ASSIGNMENT - 1

UNIT - 1

Task 1

Case	Learning paradigm	Learning task	Experience E	Performance P
(a) Loan default	• Supervised	• Binary classification	• 10 years labelled repayment records	• Accuracy, precision/recall.
(b) Customer segments	• Unsupervised	• clustering	• 2 million unlabelled customer records	• cluster quality / cohesion
(c) Credit-card spend	• Supervised	• Regression	• Historical labelled spending data	• MSE/RMSE
(d) ATM cash refill	• Reinforcement	• control / decision-making	• Repeated actions and feedback/rewards.	• Total reward/cash availability
(e) Fraud detection	• Semi-supervised	• Binary classification	• 2% labelled + 98% unlabelled transactions	• Precision, recall, F1

• Most affected by label noise.

(b) loan-default prediction, because it relies entirely on labelled repayment outcomes. Incorrect labels directly teach the classifier wrong relationships.

2. Task-2

Given:

- Prevalence = P(D) = 0.02
- Sensitivity = P(+|D) = 0.95
- Specificity = 0.90
- False-positive rate = 0.10

- Using Bayes theorem:

$$P(D|+) = \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|\bar{D})P(\bar{D})}$$

$$= \frac{0.95(0.02)}{0.95(0.02) + 0.10(0.98)}$$

$$= \frac{0.019}{0.117} = 0.1624$$

$$= P(D|+) \approx 16.24\%$$

- Although sensitivity is 95%, the disease is rare. Therefore false positives from the much larger healthy population significantly reduce the probability that a positive result is truly a disease case.

Task 3:

Joint Probabilities:

X\Y	0	1
0	0.10	0.20
1	0.15	0.25
2	0.20	0.10

1) Marginals

$P_X(0) = 0.30$, $P_X(1) = 0.40$, $P_X(2) = 0.30$

$P_Y(0) = 0.45$, $P_Y(1) = 0.55$

2) Mean and Variance:

$E[X] = 0(0.3) + 1(0.4) + 2(0.3) = 1$

$E[X^2] = 0 + 0.4 + 4(0.3) = 1.6$

4 $\text{Var}(X) = 1.6 - 1^2 = 0.6$

$E[Y] = 0(0.45) + 1(0.55) = 0.55$

3) Conditional Probability

$P(X=2|Y=1) = \frac{P(2,1)}{P(Y=1)} = \frac{0.10}{0.55} = 0.1818$

4) Covariance

$E[XY] = 0.25 + 0.20 = 0.45$

$\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$

$= 0.45 - (1)(0.55) = -0.10$

$\rho = \frac{-0.10}{\sqrt{0.6(0.2475)}} \approx -0.2595$

(c) Independence

For independence:

$P(X=0, Y=0) = P(X=0)\,P(Y=0)$

But:

$0.10 \neq (0.30)(0.45)$ $\quad 0.135$

Therefore, X and Y are not independent

The negative covariance indicates a weak negative relationship.

Task – 4

Given;

$X \sim N(50, 16)$

Therefore:

$\mu = 50, \quad \sigma = 4$

(a) Probabilities

For $X > 58$:

$Z = \frac{58 - 50}{4} = 2$

$P(X > 58) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$

For $46 < X < 54$:

$Z_1 = \frac{46 - 50}{4} = -1, \quad Z_2 = 1$

$P(46 < X < 54) = \Phi(1) - \Phi(-1)$

$= 0.8413 - 0.1587 = 0.6826$

ML estimates

- Data: 12, 15, 11, 14, 13

- Mean:

$$\hat{M}_{ML} = \frac{12+15+11+14+13}{5} = 13$$

- ML variance:

$$\hat{\sigma}^2_{ML} = \frac{\Sigma(x_i - \bar{x})^2}{n} = \frac{10}{5} = 2$$

- Unbiased variance:

$$s^2 = \frac{10}{4} = 2.5$$

- ML variance = 2, while unbiased variance = 2.5. ML variance is biased downward because it divides by n rather than $n-1$.

Task 5

- Probabilities: $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$

- Entropy:

$$H = -\Sigma P_i \log_2 P_i$$

$$= 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3)$$

$$H = 1.75 \text{ bits}$$

2

- Uniform four symbol source:

$$H_{uniform} = \log_2 4 = 2 \text{ bits}$$

- Redundancy:

$$2 - 1.75 = 0.25 \text{ bits}$$

Percentage redundancy:

$= \frac{0.25}{2} \times 100 = 12.5\%$

- Average code length for {0, 10, 110, 111}:

$L = 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3)$

$L = 1.75$ bits/symbol

- Since average length equals entropy, the code is optimal for this source.

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(6) Task 6

$P = (0.5, 0.25, 0.25)$, $Q = (1/3, 1/3, 1/3)$

KL (P||Q)

$D_{KL}(P||Q) = \sum P \log_2 \frac{P}{Q}$

$D_{KL}(P||Q) \approx 0.085$ bits

KL (Q||P)

$D_{KL}(Q||P) \approx 0.082$ bits

KL divergence is asymmetric:

$D_{KL}(P||Q) \neq D_{KL}(Q||P)$

Therefore, the choice of direction matters when KL divergence is used as a training loss.

Task 7

Given:

$P(0,0) = 0.4$ $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$

- Both A and B are equally likely:

$$P(A=0) = P(A=1) = 0.5$$

- therefore:

$$H(A) = H(B) = 1 \text{ bit}$$

- Joint Entropy:

$$H(A,B) \approx 1.722 \text{ bits}$$

- Conditional entropy:

$$H(A|B) = H(A,B) - H(B)$$

$$= 0.722 \text{ bits}$$

- Mutual information:

$$I(A;B) = H(A) - H(A|B)$$

$$= 0.278 \text{ bits}$$

- Thus B contains useful information about A, but it is not a perfect predictor.

Task 8

- Given;

$$\mu_1 = 2, \quad \mu_2 = 6, \quad \sigma^2 = 4$$

$$P(\omega_1) = 0.6, \quad P(\omega_2) = 0.4$$

- For equal variances and zero-one loss, the Baye's threshold satisfies:

$$x^* = 4 + \ln\left(\frac{0.6}{0.4}\right)$$

$$x^* \approx 4.405$$

- Classification of $x = 4.5$

Since:

$4.5 > 4.405$

$x = 4.5 \rightarrow \omega_2$

Overall error

$P_e = P(\omega_1) P(x > x^* / \omega_1) + P(\omega_2) P(x < x^* / \omega_2)$

$P_e \approx 0.154$ (15.4 %)

- If the prior probability of ω_1 increases, the threshold moves to the right, making classification of ω_1 more likely.

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3) Task 9

Given:

$P(\omega_1 | x) = 0.3, \quad P(\omega_2 | x) = 0.7$

- Risk of declaring disease

$R(\alpha_1 | x) = 0(0.3) + 8(0.7) = 5.6$

- Risk of declaring healthy

$R(\alpha_2 | x) = 50(0.3) + 0(0.7) = 15$

- Therefore:

α_1 = declare disease

is the minimum-risk action.

- MAP would select healthy, because $0.7 > 0.3$, Hence, minimum-risk decision differs from MAP because missing disease has a much larger cost

So, to be optimal:

$$8(1-p) < 50p$$

$$8 < 58p$$

$$p > 0.1379$$

So, disease should be declared when posterior probability exceeds approximately 13.79 %.

(D) Task 10

Degree	Training MSE	Validation MSE
1	9.0	9.5
2	4.1	4.6
3	2.2	2.9
5	0.9	3.6
9	0.1	11.8

Diagnosis:

Degree 1: underfitting - high training and validation error.

Degree 2: Best balance between bias and variance.

Degree 3: Beginning of overfitting.

Degree 9: Severe overfitting - very low training error but extremely high validation error

Best model: Degree 2

because it has the lowest validation MSE of 4.6.

- Improve degree-9 model:

(1) Apply regularization such as <Ridge/Lasso

(2) Reduce model complexity or use cross-validation to select a suitable degree.

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(1) Task 22

- Beta: $n = 10,\ h = 7,\ t = 3$

- Prior: $\theta \sim Beta(2, 2)$

- ML estimate:

$\hat{\theta}_{ML} = \frac{7}{10} = 0.7$

- Posterior:

$Beta(2+7, 2+3) = Beta(9, 5)$

- MAP estimate:

$\hat{\theta}_{MAP} = \frac{9-1}{9+5-2} = \frac{8}{12} = 0.667$

For 70 heads in 100 tosses:

$E[\theta|D] = \frac{9}{14} = 0.643$

- Posterior:

$\hat{\theta}_{ML} = 70/100 = 0.70$

- MAP:

$Beta(72, 32)$

Posterior mean:

$\frac{71}{102} = 0.696$

$= \frac{72}{104} = 0.692$

with more data the influence of the prior becomes smaller, so the Bayesian estimates move closer to the ML estimate.

Task 12

a) Learning System

T: Predict sepsis within the next 6 hours.

P: sensitivity, specificity, Precision and F_1 score.

E: Historical ICU EHR data.

Target: Probability of sepsis.

Inputs: vital signs and lab values.

Algorithm: Random Forest.

Validation: Stratified train test split

b) Confusion matrix

- TP = 45
- FN = 5
- FP = 190
- TN = 760.

c) PPV and NPV

$$PPV = \frac{45}{45+190} = 19.13\%$$

$$NPV = \frac{760}{760+5} = 99.35\%$$

d) Cost

Using model = $5 \times ₹100 + 190 \times ₹5 = ₹1,450$

Alert everyone = $950 \times ₹5 = ₹4,750$

Never Alert = $50 \times ₹100 = ₹5,000$

Best:

Model, because it has the lowest cost (≈ 1,450)

e) Deployment policy

- use a cost-sensitive threshold because missing sepsis costs much more than a false alarm. Alert when the predicted sepsis probability is approximately 4.76% or higher

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - I

Programme	B.E / B.Tech	**Course Code**	MLA02
Course Name	Fundamentals of Machine learning.	**Date**	31/10/26
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: 192525412

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	4
Task 4	4	4
Task 5	3	2
Task 6	3	3
Task 7	4	3
Task 8	4	2
Task 9	3	3
Task 10	4	4
Task 11	4	4
Task 12	8	6
TOTAL	**50**	42

Total in words: